

Progressive lake warming in Central Europe: six decades of trends in lake heat waves in Poland

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Abstract

Against the backdrop of observed climate changes, there is growing interest in the extreme course of processes and phenomena in the natural environment. This trend is also being explored in relation to various components of the hydrosphere, in which one fundamental characteristic is water temperature. The aim of this study was to assess the temporal and spatial variability of heat waves occurring in lakes. The research material included daily water temperatures from 1964 to 2023 for nine temperate-zone lakes with diverse morphometric parameters. The study found that the duration, cumulative intensity, and maximum intensity of heat waves increased over the last six decades. In general, the first half of the analysed period was relatively stable in the context of the study, whereas the most recent three decades showed an increase in various heat wave characteristics. A sharp increase in the duration and cumulative intensity of heat waves was observed, along with greater variability in these parameters across years and among the studied lakes. The average rate of change in heat wave duration, maximum intensity, and cumulative intensity increased by 15 days per decade, 0.5 °C per decade, and 50 °C per decade, respectively. Additionally, the changes in average heat wave duration and cumulative intensity more than doubled between 1996 and 2023. These findings should be considered unfavourable in both natural processes and the economic aspects of lake utilization. This situation creates a new reality that should be taken into account when considering potential restoration efforts to improve water quality.

Keywords: extreme hydrological events, global warming, inland water, Poland

Received March 2026, accepted September 2026.

Introduction

The response of natural environment components to observed climate changes varies due to the complexity of their interactions with the atmosphere. A clear reaction is observed in the surface water layer of lakes, where air temperature determines the thermal conditions of these ecosystems. Lake temperature is widely recognized as a fundamental property of water, influencing the frequency

and intensity of numerous processes. As a result, the study of thermal regimes attracts significant interest and holds an interdisciplinary dimension, mainly addressing two key aspects: biotic and abiotic conditions. The increase in water temperature in Lake Iseo (Italy) has been linked to higher nutrient concentrations (DORY, F. *et al.* 2024). Studies on *Escherichia coli* in lakes have revealed significant interactions affecting *E. coli* survival, particularly between temperature and to-

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tal dissolved solids, as well as temperature and coexisting microbes (YONEDA, I. *et al.* 2024). Temperature has been positively correlated with methane (CH_4) concentration in the water column, bubble emissions, and diffusion, as confirmed by studies of two polymictic lakes in Brazil (PALMA-SILVA, C. *et al.* 2013). The low abundance of large cladocera in shallow urban lakes (Lingan) has been attributed to high water temperatures, food limitations, and predator pressure (CHEN, G. 2021). The way lakes function is undergoing a gradual transformation due to climate warming and the associated rise in water temperature. Lake warming has been well documented in many regions worldwide. The surface water temperature of Lake Nam Co (Tibetan Plateau) increased significantly ($p < 0.05$) between 1963 and 2019, at a rate of 0.41°C per decade (GUO, L. *et al.* 2023). In Lake Balaton (Hungary), water temperature has been rising by 0.7°C per decade, with a more pronounced warming trend in annual minimum and winter temperatures (LI, H. *et al.* 2024). Based on data from over 500 lakes in north-western Canada, the average increase in water temperature during the open-water season has been recorded at 0.03°C per year (ATTIAH, G. *et al.* 2023).

However, increasing attention is being given to extreme events, whose occurrence may have negative environmental consequences. Unlike the rise in average annual temperature, heat waves are considered more significant for natural populations (MAMERI, D. *et al.* 2020), including those in inland waters. Regarding temperature, the greatest impact may result from increased climate variability and the number of extremely hot days and heat waves (HANDISYDE, N. *et al.* 2006). CHEN, W. *et al.* (2020) highlight the steady rise in the frequency of extreme climate events, which could have harmful consequences for lake ecosystems. Exceeding certain thermal thresholds can affect lake functioning. Changes in water mixing patterns may occur once a specific temperature threshold is surpassed (SCHMID, M. *et al.* 2014). Prolonged

above-average temperatures are linked to warming of water bodies, limiting lake overturning and potentially leading to anoxic conditions at the lake bottom (SMOYER-TOMIC, K.E. *et al.* 2003).

An analysis of physio-chemical thresholds in French lakes found that maximum water temperature plays a crucial role in fish species distribution (ROUBEIX, V. *et al.* 2017). The impact of high temperatures, alongside changes in the N:P ratio, contributed to declines in zooplankton density over time and a reduction in cladocerans relative to copepods (KELLY, P.T. *et al.* 2024). Internal loading in the shallow Littoistenjärvi lake (Finland) showed a significant positive correlation with maximum water temperature and pH (SARVALA, J. *et al.* 2020). Many salmonid fish species are at risk of thermal stress and habitat loss due to rising temperatures during summer heat waves (ADAMEK, M. *et al.* 2023). From a climate perspective, even a short-term heat wave can have widespread hydrological effects within a given year (WHITFIELD, P.H. *et al.* 2024). Therefore, in the context of existing knowledge on lake thermal regimes, a significant research trend has emerged in recent years, focusing on extreme events – specifically, lake heat waves (ZHANG, K.X. and YAO, Y.L. 2023; DUAN, Z. *et al.* 2024; WU, Y. *et al.* 2024). In Poland, lakes represent a crucial element of the natural environment, influencing not only ecological conditions but also the national economy (NOWAK, B. and DUMIŃSKI, G. 2020). Polish lakes have been studied in relation to thermal conditions, a fundamental characteristic of the hydrosphere (ZHU, S. *et al.* 2020). To date, research on heat waves in Polish lakes has analysed the 2018 event against data from 1987–2020 (SUN, J. *et al.* 2025).

Results indicate that the European heat wave of 2018 increased the mean and maximum lake surface water temperatures (LSWTs) by 2.35°C and 3.38°C , respectively, relative to the multi-year average. These findings encourage further analysis and a deeper understanding of heat wave occurrences in

Polish lakes. Detailed monitoring of water temperatures in selected Polish lakes enables the realization of such research objectives. More broadly, this study, based on sixty years of daily field measurements from nine lakes with diverse morphometric characteristics, contributes to the global understanding of lake ecosystem functioning. Against this background, the study aimed to determine the frequency and duration of lake heat waves and to identify trends in their occurrence over the past six decades. It should be emphasised that this approach aligns with the current direction of hydrological research, which increasingly focuses on extreme events. As highlighted by JAMALI, M. and ESLAMIAN, S. (2023), in the case of extreme hydrological events, trend analysis is an appropriate method for obtaining the essential practical information required for regional water resource management and planning.

Materials and methods

Study area and materials

The study uses surface water temperature data from nine lakes (Table 1) located in the lake districts of northern Poland (Figure 1).

The measurements were systematically collected by the Institute of Meteorology and Water Management (IMGW). The dataset comprises detailed daily water temperature records spanning six decades (1964–2023). Measurements were taken at a fixed location in each lake, at a depth of 0.4 m, every morning at 6:00 UTC. The study also uses daily air temperature data collected by the same institute over the same period in accordance with standard meteorological measurement procedures.

Methods

The methodology proposed by HOBDDAY, A.J. et al. (2016) was used to identify lake water heat waves (LHW). An anomalously warm event lasting at least five consecutive days, during which water temperature exceeded the threshold, was classified as an LHW. Threshold values for each lake for each day of the year were defined based on the 90th percentile values from a 30-year historical baseline period. In this study, 1964–1993 was adopted as the baseline period. Events separated by gaps of two days or fewer and subsequent heat wave events of five days or more were treated as a single continuous heat wave event. For each LHW, the start date, end

Table 1. Lakes morphometric parameters

No.	Lake	Latitude North (Lake)	Longitude East (Lake)	Depth*		Area,* ha	Altitude,* m a.s.l.	Volume,* m ³
				max, m	mean, m			
1	Ślowskie	51.89	16.02	12.3	5.2	822.5	56.9	42,664.80
2	Gardno	54.71	17.39	2.6	1.3	2337.5	0.3	30,950.50
3	Łebsko	54.65	17.10	6.3	1.6	7020.0	0.3	117,521.00
4	Charzykowskie	53.77	17.50	30.5	9.8	1336.0	120.0	134,533.20
5	Jeziorak	53.72	19.62	12.9	4.1	3152.5	99.2	141,594.20
6	Mikołajskie	53.77	21.60	25.9	11.2	424.0	115.7	55,739.70
7	Hańcza	54.27	22.81	106.1	38.7	291.5	227.3	120,364.10
8	Białe Augustowskie	53.86	23.03	30.0	8.7	476.6	122.0	41,700.00
9	Studzieniczne	53.87	23.12	30.5	8.7	244.0	123.4	22,073.60

*After CHOŃSKI, A. 2006.

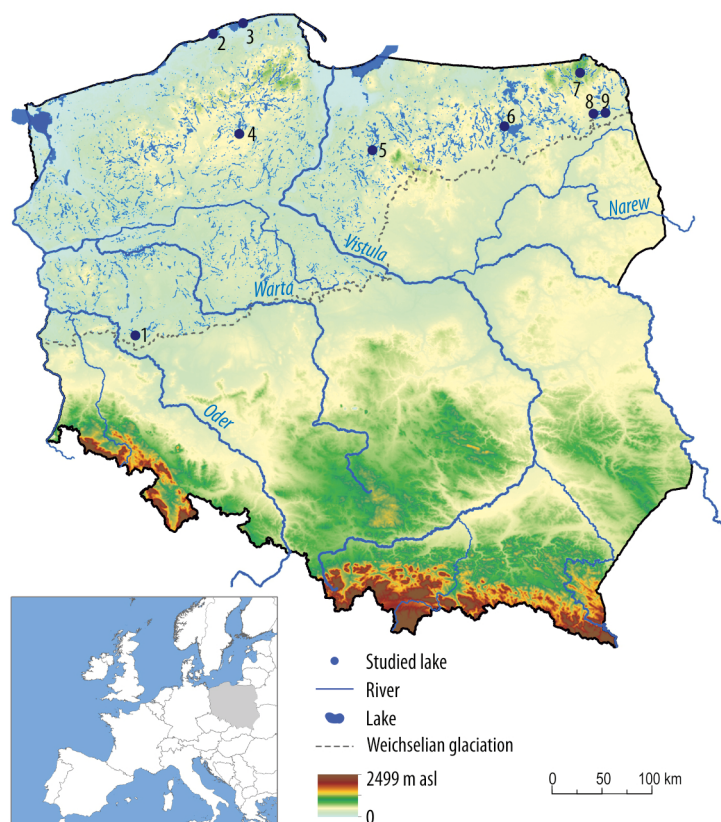


Fig. 1. Location of studied lake inside the territory of Poland. Lakes: 1 = Sławskie; 2 = Gardno; 3 = Łebsko; 4 = Charzykowski; 5 = Jeziorak; 6 = Mikołajskie; 7 = Hańcza; 8 = Białe Augustowskie; 9 = Studzienne. Source: Authors' own elaboration.

date, duration (DT), mean intensity (IMean), maximum intensity (IMax), and cumulative intensity (CI) were determined. Annual statistical summaries of these parameters were subsequently prepared. Where an LHW spanned two consecutive calendar years, it was divided into two separate episodes to enable the preparation of annual summaries. In addition, LHW were classified into four categories – moderate, strong, severe, and extreme – according to their maximum intensity, following the methodology of HOBDAI, A.J. *et al.* (2018). Statistical summaries for each heat wave category were prepared for six decadal periods: 1964–1973, 1974–1983, 1984–1993, 1994–2003, 2004–2013, and 2014–2023.

The resulting values were further analysed to identify long-term changes in lake water heat wave parameters between 1964 and 2023. In the first stage of the analysis, the non-parametric Pettitt test (PETTITT, A.N. 1979) was used to identify break points in the annual DT, IMean, IMax, and CT series from the period 1964–2023. The non-parametric Mann-Kendall test (KENDALL, M.G. and STUART, A. 1968), and Sen test (GILBERT, R.O. 1987) were then used to analyse long-term changes in DT, IMean, IMax, and CI values in the study lakes. Two hypotheses were formulated for the Mann-Kendall trend analysis: the null hypothesis (H_0) that no trend is present in the time series, and the

alternative hypothesis (H1) that a trend is present. The magnitude of the identified trends in annual DT, IMean, IMax and CI values was estimated using Sen's non-parametric test. Before the trend analysis, the time series were tested for autocorrelation and, where necessary, its influence was removed (YUE, S. *et al.* 2002). The R package RmarineHeat waves, developed by SMIT, A.J. and co-authors (2018), was used to identify lake-water heat waves. The Mann-Kendall's and Sen's tests were applied using the modified R package mk developed by PATAKAMURI, S.K. and O'BRIEN, N. (2022). Finally, principal component analysis (PCA) was performed to identify the location and lake morphometric variables associated with changes in DT, IMean, IMax, and CI. The PCA analysis was conducted using Statistica 13.1 software.

Results

The results showed that lake heat waves occurred in nearly every year of the analysed 60-year period. In Lake Ślaskie (#1), heat waves were recorded in 54 years, whereas in Lake Gardno (#2), in 58 years. In 46 years of the 60 years, lake heat waves occurred in all nine lakes (Figure 2).

Across the nine lakes, the mean cumulative duration of lake heat waves was ranging from 1,927 days (8.8%) in Lake Łebsko (#3) to 3,510 days (16.0%) in Lake Charzykowskie (#4). Among individual years, 2020 was characterised by the greatest lake heat wave activity across all nine lakes (Figure 3). In that year, the mean cumulative heat wave duration was 176 days per lake, ranging from 79 to 269 days in Lake Łebs-

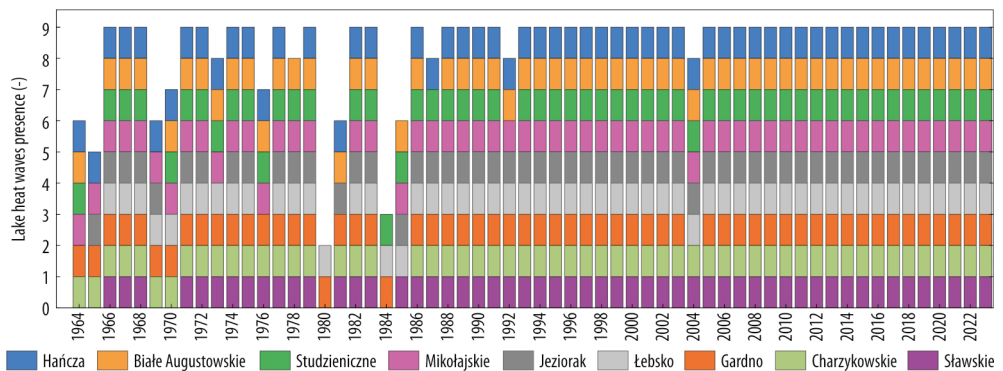


Fig. 2. Heat waves occurrence in the studied lakes. Source: Authors' own elaboration.

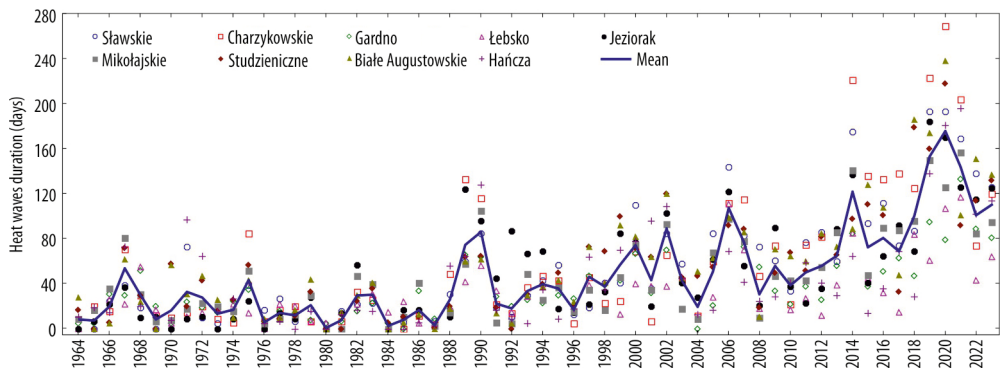


Fig. 3. Duration of lake water heat waves in the period 1964–2023. Source: Authors' own elaboration.

sko (#3) and Lake Charzykowskie (#4), respectively.

The mean annual cumulative LHW intensity across all the analysed lakes was 159 °C per year, ranging from 126 °C to 180 °C per year among individual lakes. Notably, cumulative LHW intensity generally reached its highest values during the final three decades of the study period, which were also characterized by the greatest variability (Figure 4). Annual IMax reached its highest values in 2000 and its lowest values in 1980 (Figure 5, a). Variability in annual IMax among the analysed lakes was greater during the first three decades of the study period, from 1964 to 1993. IMean exhibited lower inter-annual variability than IMax (Figure 5, b). As with IMax, variability in IMean among the lakes was greater during the first three decades of the analysed period.

Another important characteristic of lake heat waves is their severity. Across all the analysed lakes, moderate heat waves accounted for approximately 86 percent of all events, while strong and severe heat waves represented about 13 percent and just over 1 percent, respectively (Figure 6). In the coastal lakes (Gardno #2, and Łebsko #3), moderate LHWs accounted for 92.6 and 91.3 percent,

respectively. Strong LHWs accounted for 7.3 and 8.7 percent of events, respectively, while no severe and extreme events were recorded in either lake. In the remaining lakes, the proportion of moderate LHWs ranged from 83.0 to 86.8 percent, that of strong LHWs ranged from 11.7 to 15.5 percent, and that of severe LHWs ranged from 0.4 to 2.3 percent. Short extreme LHW events were also recorded, particularly in lakes located in eastern Poland (Hańcza #7, Białe Augustowskie #8, and Studzieniczne #9), 0.1, 0.1 and 0.2 percent, respectively.

Analysis of the variability of heat wave categories across the six decadal periods 1964–1973, 1974–1983, 1984–1993, 1994–2003, 2004–2013, and 2014–2023 showed that strong and severe LHWs were already present during the first decade, accounting for 7.3 and 2.3 percent of all events, respectively (Figure 7). In the period 1974–1983, LHWs were the least severe, with 97 percent of moderate category LHWs and about 3 percent of strong category LHWs. Over the following four decades, the proportion of strong LHWs increased progressively to 6.9, 10.4, 11.8, and 17.9 percent respectively, while the proportion of moderate LHWs decreased from 93.0 to 80.8 percent. The proportion of

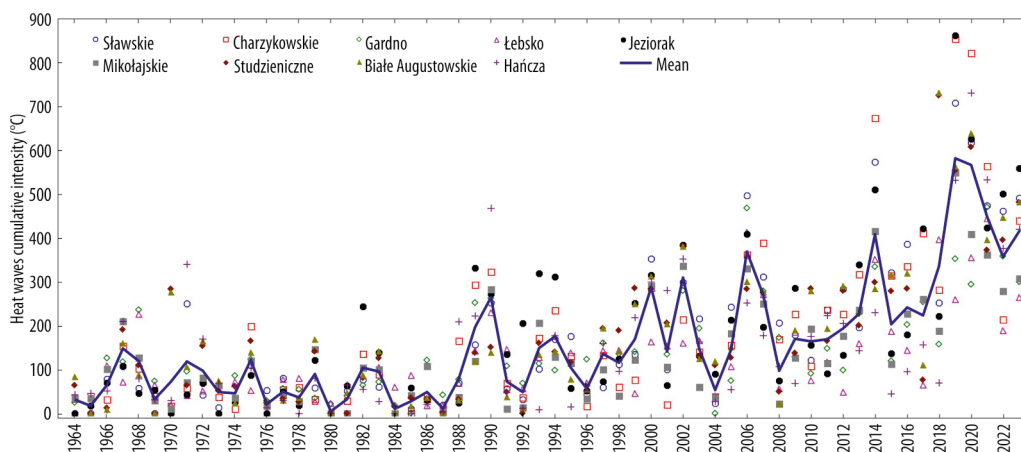


Fig. 4. Variability of cumulative intensities of lake water heat waves from 1964 to 2023. Source: Authors' own elaboration.

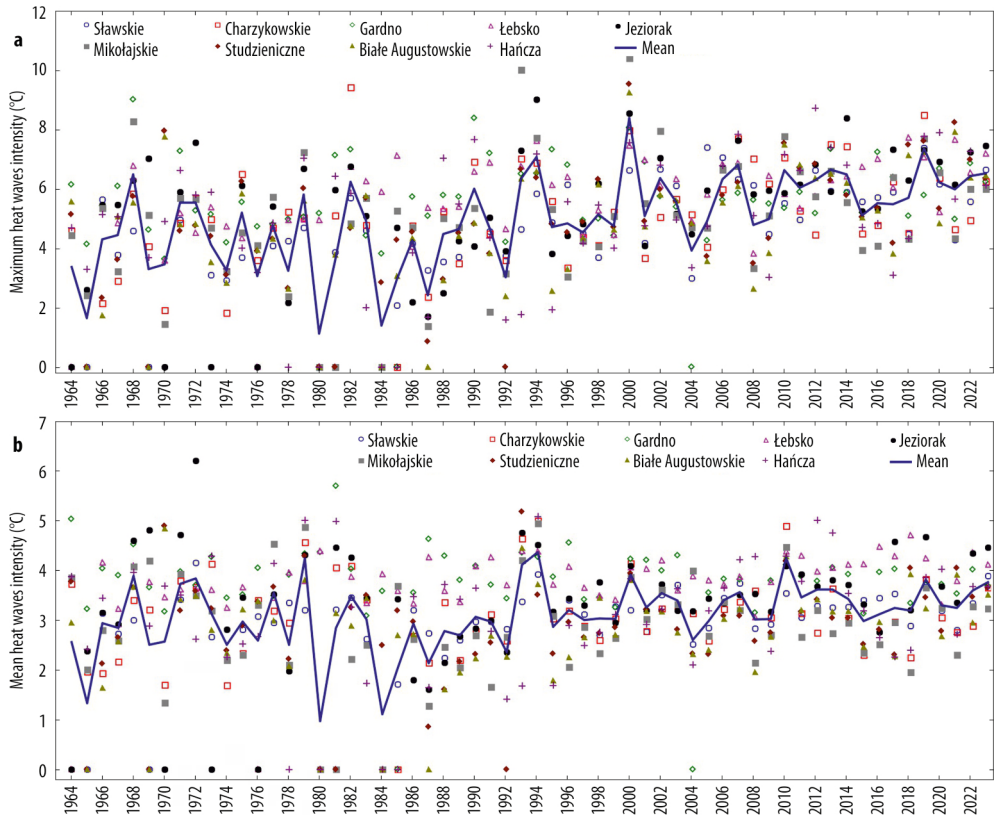


Fig. 5. Variability of maximum (a), and mean (b) annual lake water heat waves intensities from 1964 to 2023. Source: Authors' own elaboration.

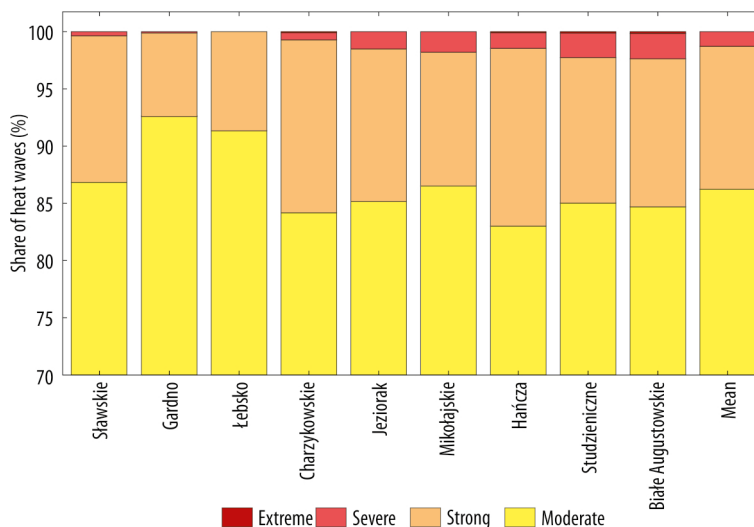


Fig. 6. Shares of different lake water heat wave categories. Source: Authors' own elaboration.

severe LHWs during the final four decades ranged from 0.1 to 2.3 per-cent, reaching its highest value in the period 2004–2013, while extreme LHWs occurred only sporadically (see Figure 7).

The decadal variability of heat-wave category events in individual lakes was spatially represented using pie charts. The diameter of each chart represents the mean annual number of LHW days, while the colours indicate the proportions of the individual LHW categories (Figure 8).

The analysis of lake heat wave char-acteristics over the past six decades revealed significant upward trends in both total DT and CI at the 0.01 significance level for all the lakes (Table 2). The smallest increases in DT were observed in the two coastal

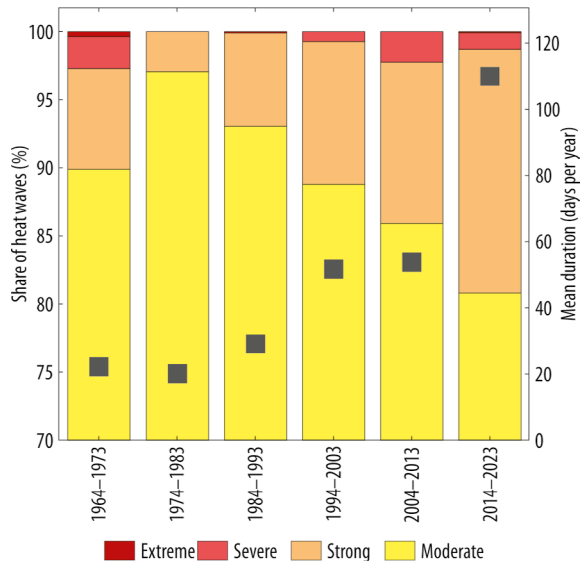


Fig. 7. Share of each category of lake water heat waves in the six decadal periods (1964–1973, 1974–1983, 1984–1993, 1994–2003, 2004–2013 and 2014–2023). Source: Authors’ own elaboration.

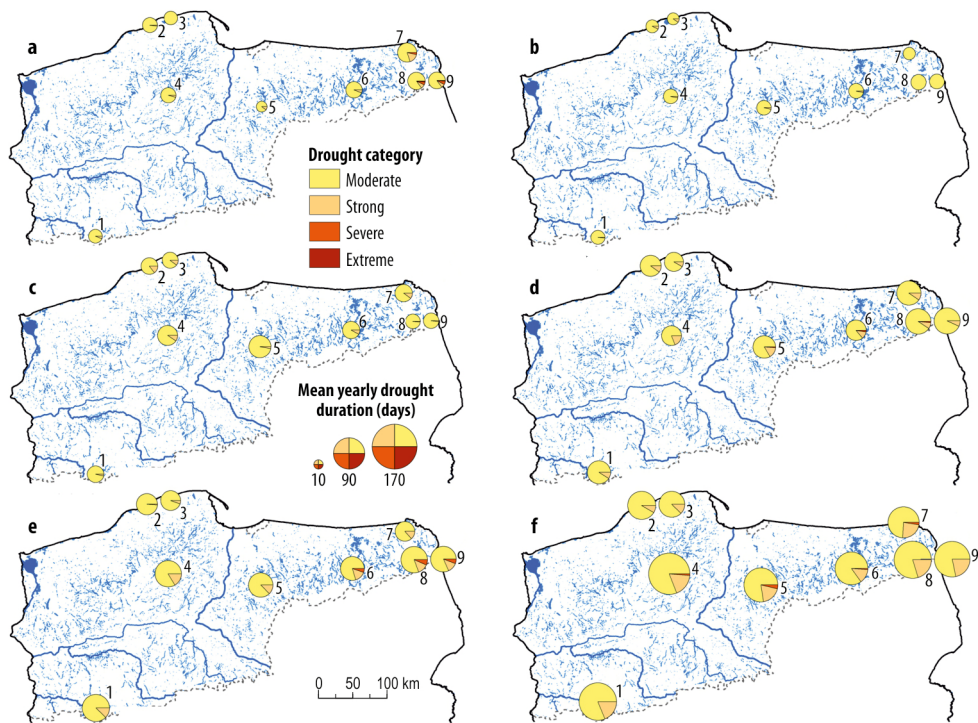


Fig. 8. Share of lake heat wave categories for the six decadal periods: 1964–1973 (a), 1974–1983 (b), 1984–1993 (c), 1994–2003 (d), 2004–2013 (e), and 2014–2023 (f). Source: Authors’ own elaboration.

Table 2. Long-term changes of the parameters describing the lake heat waves

Lake	Parameter	S-value	Tau	Z-value	p-value	Sen's slope value
Sławskie	DT	993	0.580	6.49	0.000	19.2
	IMax	719	0.420	4.70	0.000	0.5
	IMean	451	0.264	2.94	0.003	0.1
	CI	1027	0.600	6.71	0.000	63.5
Gardno	DT	853	0.499	5.57	0.000	9.6
	IMax	259	0.151	1.69	0.092	–
	IMean	–109	–0.064	–0.71	0.480	–
	CI	839	0.490	5.48	0.000	36.1
Łebsko	DT	735	0.430	4.80	0.000	7.9
	IMax	627	0.366	4.09	0.000	0.4
	IMean	481	0.281	3.14	0.002	0.1
	CI	781	0.456	5.10	0.000	33.3
Charzykowskie	DT	913	0.534	5.96	0.000	20.6
	IMax	499	0.292	3.26	0.001	0.4
	IMean	107	0.063	0.69	0.488	–
	CI	911	0.532	5.95	0.000	58.4
Jeziorak	DT	917	0.536	5.99	0.000	15.8
	IMax	571	0.334	3.73	0.000	0.7
	IMean	407	0.238	2.66	0.008	0.3
	CI	903	0.528	5.90	0.000	58.8
Mikołajskie	DT	775	0.453	5.06	0.000	14.3
	IMax	335	0.196	2.18	0.029	0.3
	IMean	143	0.084	0.93	0.353	–
	CI	703	0.411	4.59	0.000	40.3
Hańcza	DT	631	0.369	4.12	0.000	9.6
	IMax	363	0.212	2.37	0.018	0.4
	IMean	237	0.139	1.54	0.123	–
	CI	667	0.390	4.36	0.000	34.7
Białe Augustowskie	DT	971	0.568	6.34	0.000	19.1
	IMax	561	0.328	3.66	0.000	0.6
	IMean	247	0.144	1.61	0.108	–
	CI	977	0.571	6.38	0.000	63.9
Studzieniczne	DT	935	0.546	6.11	0.000	18.1
	IMax	635	0.371	4.15	0.000	0.6
	IMean	245	0.143	1.60	0.111	–
	CI	919	0.537	6.00	0.000	57.3
Mean	DT	987	0.577	6.45	0.000	14.9
	IMax	709	0.414	4.63	0.000	0.5
	IMean	498	0.291	3.25	0.001	0.1
	CI	995	0.582	6.50	0.000	49.5

Source: Authors' own elaboration.

lakes and Lake Hańcza. In the remaining six lakes, DT increased by an average of 15 days per decade. This increase was accompanied by a corresponding rise in cumulative intensity, with CI increasing at rates ranging from 33 °C to 64 °C per decade. The smallest increases were again recorded in the two coastal lakes and Lake Hańcza, whereas the remaining six lakes exhibited higher rates of increase, averaging 49 °C per decade. Maximum heat wave intensity (IMax) also increased in all studied lakes, at an average rate of 0.5 °C per decade.

Principal component statistical analysis showed that the mean duration and average cumulative intensity of heat waves increased with distance from the Baltic Sea (Figure 9). Furthermore, both maximum and average intensity of heat waves were higher in larger lakes and increased with latitude.

Discussion

The occurrence and impacts of extreme natural events and processes are an increasingly prominent topic in environmental sciences, as evidenced, among others, by hydrologi-

cal research (REICH, P. and LAKE, S. 2015; DE LUCA, P. *et al.* 2020; ASHRAF, M.S. *et al.* 2023). In line with this trend, a study was conducted on lake heat waves in Central Europe over the past six decades. In general, the first half of the analysed period was relatively stable in the context of the research, while the second thirty-year period showed an increase in various characteristics of heat waves. This period is associated with a shift in the climate regime, which, in lakes, has been confirmed by rising water temperatures in Central Europe since the late 1980s (WOOLWAY, R.I. *et al.* 2017).

The characteristics of heat waves varied between individual lakes. As shown by statistical analysis, their average duration and cumulative intensity increased with greater distance from the Baltic Sea (see Figure 9). The moderating influence of the sea was also evident in long-term changes in heat wave duration, with the smallest increases observed in coastal lakes such as Łebsko (#3) and Gardno (#2), as well as in Lake Hańcza (#7) (8 and 10 days per decade). The first two lakes not only experience maritime influences on atmospheric conditions but also have a hydraulic connection to the sea, where marine intrusions affect the physical and chemical properties of lake water. Lake Hańcza (#7), on the other hand, is the deepest lake in the Central European Plain, and as previous studies have shown (PTAK, M. and SOJKA, M. 2021), its morphometric characteristics significantly modify the impact of overarching factors such as climate conditions. This is evident in the lake's thermal behaviour, where noticeable changes in surface water temperature occurred about a decade later than in other lakes in this part of Europe. Further results indicate that the share of intense heat waves is highest among lakes in eastern Poland (the Masurian Lake District). Due to the dominant influence of the continental climate, this area historically retained ice cover the longest, effectively isolating the water from external factors. However, this state is clearly changing, and

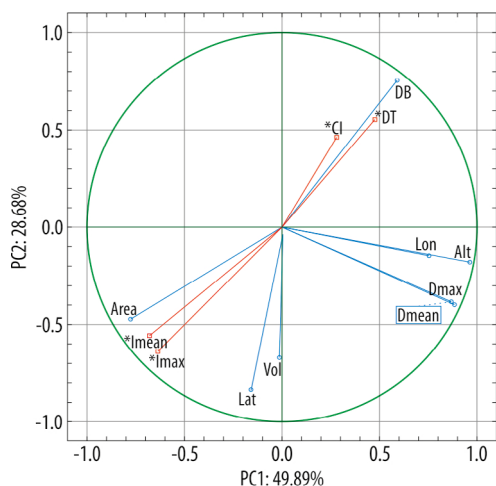


Fig. 9. Results of principal component analysis between heat waves parameters and location/morphometric lakes. Source: Authors' own elaboration.

the ice season has become significantly shorter, especially in recent decades (PTAK, M. and SOJKA, M. 2021). In extreme cases, ice cover did not form at all, for instance, in 2020. Therefore, the lakes undergoing the greatest transformation in their functioning are those that, due to the presence of ice for several months each year, were previously well insulated from atmospheric influences, but this period of insulation is now decreasing significantly.

The increase in heat waves in Europe may impact water quality and the ecological functioning of aquatic systems (FREE, G. *et al.* 2021). In terms of water quality, according to the classification of uniform bodies of surface water in line with the Water Framework Directive guidelines, all lakes (for which relevant data are available) are characterized by a poor overall status (isok.gov.pl/hydroportal.html). Moreover, the catchments of all these lakes (for which data have been compiled) are sensitive to eutrophication, which accelerates the growth of algae and higher forms of aquatic vegetation. This leads to undesirable disturbances in the biological balance of aquatic environments and a deterioration in water quality.

Among the most harmful effects of eutrophication are toxic cyanobacterial algae blooms (TOPOROWSKA, M. and PAWLIK-SKOWROŃSKA, B. 2011), and such events have been observed in the lakes analysed. The massive growth of cyanobacteria is dependent on anthropogenic pressures and rising water temperatures.

The observed increase in the frequency and duration of heat waves requires that this information be taken into account in lake restoration strategies and the pursuit of specific water quality standards. The selection of a potential restoration method should account for extreme conditions that may significantly compromise the intended outcomes of the interventions. Beyond technical solutions, water possesses natural properties that influence its self-purification ability.

One of the key parameters for the functioning of lake ecosystems is the amount of dissolved oxygen, which will decrease as the water temperature rises (PTAK, M. and NOWAK, B. 2016). Based on monitoring studies conducted in Poland by the Chief Inspectorate of Environmental Protection (wody.gios.gov.pl/pjwp), it can be stated that for selected lakes analysed in this study, oxygen saturation in the surface water layer is high. This reflects processes such as photosynthesis occurring in these waters. In deep lakes, however, the occurrence of heat waves intensifies thermal stratification, which has further consequences. Research by STETLER, J.T. *et al.* (2021) revealed a shallowing of the stratification depth in Crater Lake, which may have significant ecological implications, particularly for stationary primary producers and by reducing nutrient upwelling. For the studied lakes, oxygen levels drop rapidly below the epilimnion, reaching very low values. For example, in the summer season (August), oxygen saturation in Lake Charzykowskie (#4) was 2 percent (at a depth of 6 m), in Lake Mikołajskie (#6) 2.3 percent (depth 8 m), and in Lake Studzieniczne (#9) 3 percent (depth 7 m).

Conclusions

Water temperature is a fundamental characteristic of lake ecosystems that influences both their biotic and abiotic functioning. In addition to understanding typical temperature patterns, it is essential to investigate extreme events, whose frequency is increasing as climate change progresses. The results of this study revealed increases in the duration, maximum intensity, and cumulative intensity of lake heat waves at rates of 15 days per decade, 0.5 °C per decade, and 50 °C per decade, respectively. Moreover, the pattern of lake heat waves has changed, with an increase in the proportion of strong events. The intensification of extreme thermal condi-

tions may affect stratification stability, oxygen conditions, the rates of biogeochemical processes, and the functioning of aquatic organisms, thereby altering the functioning of entire lake ecosystems. This is therefore a complex mechanism that requires an interdisciplinary approach, including in practical management and conservation measures. These emerging conditions should be considered when planning and evaluating the effectiveness of future protection and restoration measures aimed at stabilising and improving the functioning of lake environments.

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